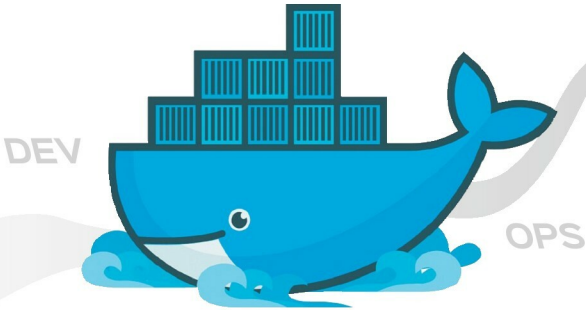


Go the DevOps Way with Docker

DevOps is a term that's emerged from 'development and operations'. It refers to the collaboration, communication and integration between software developer and IT operations teams, leading to faster production of software and services. This tutorial starts off using the official MySQL image with a minimal Dockerfile, and then moves to Docker Compose to set up a development environment for Rails.



Of late, Docker has clearly emerged as a winner in the DevOps world. It is not a panacea for all problems, though, and has a few issues, the biggest being the learning curve. Docker is fairly new while the technology it uses, i.e., Linux containers, has been around for well over 10 years. Google is a pioneer in the development of Linux containers, while Docker has come up with a more user-friendly approach to using containers.

Let's consider an example of a Rails application that uses MySQL and Redis. If you are running DevOps for your startup, one of your roles would be to bring on board new developers and help them set up services on their machines. You will probably use some automation tool like Vagrant to get this done. If you are doing it manually via `apt-get` or `Brew`, then this is the best time to start using Docker. Dockerising a service simply means adding a 'Dockerfile' to your project, just like a `Vagrantfile`. In this tutorial, we will start with using the official MySQL image with a minimal Dockerfile, and later move to Docker Compose to set up a development environment for Rails. This article assumes readers have a basic knowledge of the Linux command line.

You must have Docker and Docker Compose installed on your machine before starting, for which the official documentation can be followed (Docker: <https://docs.docker.com/engine/installation>; Docker-compose: <https://docs.docker.com/compose/install>).

Once you have Docker installed, you can verify it by running `docker` without any arguments. You can verify your installation as shown below:

```
<code: shell>
$ docker -v
Docker version 1.12.3

$ docker-compose -v
docker-compose version 1.8.0-rc2, build c72c966
```

Docker requires `sudo` to run. However, we will not use Docker with `sudo`, because we wish to avoid repetition and maintain simplicity. You can copy-paste the commands shown below to avoid using `sudo`. (Logout → Login for group changes to take effect.)